Group Participation Sheet

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A portion of this course is devoted to group work. This **Group Participation Sheet** (GPS), hereinafter referred to as the **GPS Sheet**, will specifically indicate the group members that actively contributed to the assignment, lab project, deliverable, or report. All names indicated on this GPS sheet will receive the same grade for the attached group work. Group members with names and signatures not shown on this GPS sheet will receive a grade of "**zero**" for the mentioned assessment. Any concerns regarding this matter should be brought to the professor's attention by the team leader prior to the submission deadline.

Please fill the table below electronically. Group members that participated in this evaluation should **write their names** and append their **signature**. The signed GPS sheet should be added as the **first page of all group evaluations**. This should be **applied** to all evaluations uploaded on SLATE as **softcopy**, and printed/stapled and presented as **hardcopy** to your professor.

Clear Form Print Form

Course Title:	INFO-53883 - AI/ML Capstone Project	
Evaluation Name:	INFO-53883 - D2 (15%, Group Work, Project Design Doc.)	
Team Number:	INFO-53883 (AI/ML Capstone) - Team 2B	
Referencing and Citation Style:	*APA Style* ○ *IEEE Style* ● *N/A* ○	
Group Members:	*First Name and Family Name of group members:*	*Signature of group members:*
	(1) Afolabi Adesina (team leader)	Afolabi Adesina Digitally signed by Afolabi Adesina Date: 2026.06.03 19:38:36 -04'00'
	(2) Evans Frimpong	Evans Frimpong
	(3) Xian Qin	Xian Qin
	(4) Yanan Yang	Yanan Yang Digitally signed by Yanan Yang Date: 2026.06.03 17:41:45 -04'00'
	(5)	
Submission Date:		

Grade:	/15

AI/ML Capstone Project

D2 Project Design Document (15% of final mark)

Due: Week 4 (softcopy on SLATE, hardcopy to the professor in class)

Name 1: Afolabi Adesina	**Student Number 1:** 991808102
Name 2: Evans Frimpong	**Student Number 2:** 991771602
Name 3: Xian Qin	**Student Number 3:** 991578381
Name 4: Yanan Yang	**Student Number 4:** 991841037

Instructions:

- Please work on the following deliverable with the group.
- Prepare the deliverable report as discussed in class.
- Type the report using word processing software (preferably using LaTeX, otherwise MS Word is fine).
- Upload the report in PDF format on SLATE. The team leader should upload this deliverable on SLATE as a single PDF file.
- Present a printout (stapled) of the deliverable to the professor in class in Week 4.

Overall Objectives:

In this D2 Project Design Document, the objective is to design a realistic and achievable project plan that defines technical architecture, milestones, and team deliverables using industry standard practices. This document will serve as a foundational blueprint for all subsequent development phases, and it must be structured with clarity and professionalism.

Q1: Project Objectives

Create a Work Breakdown Structure (WBS) detailing technical tasks and deliverables. As part of the WBS, include at least four Work Packages (WPs). These WPs should address the major objectives of the AI/ML system architecture and data flow intended for this capstone project. In the space below, provide an explicit heading for each WP followed by a brief description detailing what exactly will be accomplished (in 4 to 6 sentences). [2 marks]

WP1: Data Gathering and Preparation (Data Layer)

- **Task 1.1: API & Live Data Integration**, Establish connections to Google Directions API, Open Meteo REST API, weather data from Env Canada, and Ontario 511 system for real time ingestion.
- **Task 1.2: Historical Data Engineering**, Download, clean, and structure the 768,000+ rows from the Toronto Police and ORSAR historical collision databases.
- **Task 1.3: Unstructured Vision Preprocessing**, Build an automated pipeline to extract frames from MTO traffic cameras and run redaction scripts to mask PII (faces and license plates).
- **Task 1.4: Unstructured Text Preprocessing**, Implement NLTK pipelines to tokenize and clean official government emergency alerts.

WP2: Machine Learning Model Creation (Framework & Model Layer)

- **Task 2.1: Vision Brain (CNN) Development**, Train a Convolutional Neural Network using PyTorch and OpenCV to classify road surface conditions into dry, wet, snowy, and icy.
- **Task 2.2: NLP Brain Development**, Apply TF IDF weighting to the cleaned text alerts to generate a quantitative Textual Risk Index.
- **Task 2.3: Logistic Optimizer (Fusion Model)**, Build a Lasso regularized Logistic Regression model using Scikit Learn (SAGA solver) to fuse visual, textual, and environmental data.
- **Task 2.4: Hyperparameter Tuning & Validation**, Utilize GridSearchCV to optimize the regularization constant, prioritizing a recall rate greater than 80% on the HIGH RISK class.

WP3: Backend System Building (Infrastructure & MLOps Layer)

- **Task 3.1: Environment Configuration**, Set up Python 3.10 virtual environments and configure local CUDA enabled GPUs or cloud instances for model training.

- **Task 3.2: Asynchronous Pipeline Automation**, Configure background cron jobs to continuously fetch external features and calculate predictive risk scores without latency.
- **Task 3.3: Containerization**, Package the backend processing scripts into Docker containers to ensure consistent execution across different environments.

WP4: Frontend Application Design (Application Layer)

- **Task 4.1: UI/UX Wireframing**, Design the layout for the standalone web application, mapping out user input fields (origin and destination) and output displays.
- **Task 4.2: Streamlit Dashboard Development**, Build the interactive web application interface utilizing Streamlit.
- **Task 4.3: Dynamic Map Integration**, Embed mapping functionality that displays ranked routes color coded by the dynamic Composite Safety Score.

Q2: Project Tasks

The focus here is on identifying the granularity of the WPs. For each WP, develop at least 3 to 4 tasks detailing the various subcomponents related to the AI/ML pipeline and applications. In the space below, provide a descriptive heading for each task. Then, for each labelled task, provide an explanation of what the task entails (in 1 to 2 sentences). Avoid being too redundant to what was indicated in Q1. [2 marks]

WP1: Data Gathering and Preparation

- 1.1 API & Live Data Integration
- 1.2 Historical Data Engineering
- 1.3 Unstructured Vision Preprocessing
- 1.4 Unstructured Text Preprocessing

WP2: Machine Learning Model Creation

- 2.1 Vision Brain (CNN) Development
- 2.2 NLP Brain Development
- 2.3 Logistic Optimizer (Fusion Model)
- 2.4 Hyperparameter Tuning & Validation

WP3: Backend System Building

- 3.1 Environment Configuration
- 3.2 Asynchronous Pipeline Automation
- 3.3 Containerization

WP4: Frontend Application Design

- 4.1 UI/UX Wireframing
- 4.2 Streamlit Dashboard Development
- 4.3 Dynamic Map Integration

Q3: Task Owners

Assign roles and responsibilities for each team member. In other words, assign a task owner for each task within the WPs. This must be done strategically based on qualifications and interest of the team member being assigned a task. Common roles include project manager, data engineer, AI/ML engineer, MLOps engineer, etc. Divide the workload equally among the team members so that it is fair, balanced, and realistic. In the space below, list the tasks from Q2, and indicate the name of the team member that will work on the specific task in parenthesis. Also, provide a specific description (in 1 to 2 sentences) why this specific team member is appropriate to work on the assigned task. [1 mark]

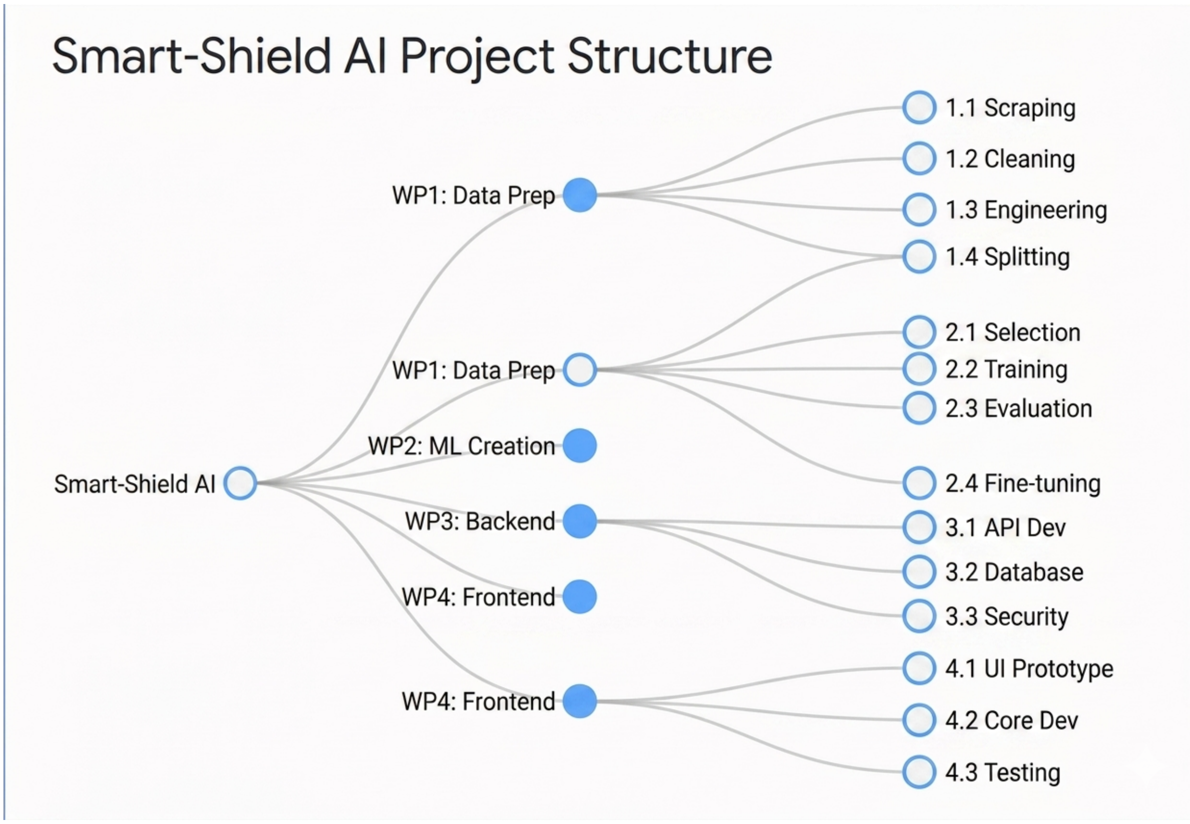
Team Roles and Responsibilities

- **Evans (Data Engineer and Backend Specialist):** Evans manages the initial data acquisition and system setup. Responsibilities include gathering historical data and collaborating on live API data integration. Evans also configures the initial backend environment to ensure a stable foundation for the project.
- **Xian (Full Stack Data Developer):** Xian handles both data ingestion and interactive frontend tasks. Responsibilities involve jointly managing live API data with Evans. Xian takes full ownership of building the Streamlit application and integrating map features for the user interface.
- **Yanan (Machine Learning and DevOps):** Yanan bridges model development and system deployment. Responsibilities start with preprocessing text and vision data. Yanan jointly develops the core Convolutional Neural Network and Natural Language Processing models. Additional duties include building asynchronous automation, managing containerization, and ensuring efficient team workflows to avoid duplicate efforts.
- **Afolabi (AI Optimizer and User Experience Designer):** Afolabi focuses on model refinement and user experience planning. Responsibilities include jointly developing the vision and text models. Afolabi leads the fusion optimizer and hyperparameter tuning phases. Afolabi also creates the initial wireframes to guide the frontend development.

All team members participate in the final Quality Assurance, testing, and project polish phase.

Q4: WBS Diagram

Using the descriptions in Q1 (work packages), Q2 (tasks), and Q3 (task owner) organize the WBS to visually capture relevant information by way of a tree diagram. The diagram should show the hierarchical relationships among all WPs and tasks, and it should include all relevant sub branches. Also, make the diagram complete and coherent by enumerating the topics and by placing arrows to show all interconnected associations. The diagram should visually display an overall summary for the workflow for the AI/ML project. Specialized software can be used to make the diagram (e.g., Visio, draw.io, photoshop, etc.). [1 mark]



Q5: Weekly Milestones

Define weekly milestones and deliverables for project tracking. The weekly milestones plan should cover Week 5 to Week 14 of the semester. Each week, a certain aspect of the project work needs to be completed (e.g., data gathering and cleaning, AI/ML modeling, model optimization, testing and prototyping, deployment and UI, tech report writing, final presentation and demonstration). In the space below, enumerate the weeks from Week 5 to Week 14, and provide a bulleted list of the tasks and sub tasks that the group aims to realistically accomplish by the end of each week. [2 marks]

- **Weeks 5–6: Data Architecture & Pre-Processing**
  - APIs connected.
  - Historical data ingested and cleaned.
  - Image redaction scripts validated.
  - Baseline environment established.

- **Weeks 7–8: Independent Model Training**
  - NLP Textual Risk Index operational.
  - Baseline CNN model classifying road conditions.
- **Weeks 9–10: Model Fusion & Backend Integration**
  - Three-brain fusion completed.
  - Composite Safety Score generated successfully.
- **Weeks 11–12: Frontend Development & Containerization**
  - Streamlit prototype launched.
  - Dynamic route mapping implemented.
  - Docker containers built.
- **Weeks 13–14: Optimization, Testing & Final Polish**
  - Model achieves greater than 90% recall on high-risk test data.
  - Documentation finalized.
  - GitHub repository cleaned.
  - Final presentation.

Q6: Gantt Chart

Using the results in Q4 (WBS diagram) and Q5 (weekly milestones), develop a detailed Gantt chart. The Gantt chart is a project management tool that visually illustrates a project schedule over time. The Gantt chart should display a list of WPs and tasks vertically while being mapped to specific task owners. A timeline is shown horizontally using horizontal bars that should be used to represent the needed duration in weeks. Make the Gantt chart as detailed as possible. Also, important AI/ML milestones should explicitly be indicted on the Gantt chart. An example of a Gantt chart that can be replicated is shown in Appendix A. [2 marks]

WBS	Task Description	Owner	Start	Due	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
WP1	Data Gathering & Prep	Evans / Xian	W5	W6	■	■								
1.1	API & Live Data	Evans / Xian	W5	W5	■	■								
1.2	Historical Data	Evans	W5	W6	■	■								
1.3	Vision Pre-Processing	Yanan	W6	W6	■	■								
1.4	Text Pre-Processing	Yanan	W6	W6	■	■								
WP2	ML Model Creation	Yanan / Afolabi	W7	W10			■	■	■	■				
2.1	Vision Brain (CNN)	Yanan / Afolabi	W7	W8			■	■						
2.2	NLP Brain Dev	Yanan / Afolabi	W7	W8			■	■						
2.3	Fusion Optimizer	Afolabi	W9	W9					■	■				
2.4	Hyperparameter Tuning	Afolabi	W10	W10					■	■				
WP3	Backend System	Xian / Evans/Yanan	W9	W12					■	■	■	■		
3.1	Env Configuration	Evans	W5	W5	■	■								
3.2	Async Automation	Xian	W9	W10					■	■				
3.3	Containerization	Xian	W11	W12							■	■		
WP4	Frontend Application	Evans	W11	W14							■	■	■	■
4.1	UI/UX Wireframing	Evans	W11	W11							■	■		
4.2	HTML, CSS, Javascript	Evans	W11	W12							■	■		
4.3	Map Integration	Evans	W12	W12							■	■		
QA	Testing & Polish	Team	W13	W14									■	■

Q7: Version Control

Establish version control practices and repository management protocols with Git/GitHub. This tool will allow tracking changes in source code and related files over the span of the AI/ML capstone project. This tool will also allow practical collaboration with team members without overwriting each other's work. In the space below, explain how Git/GitHub will be used for version control and for repository planning. Also describe in full detail the collaboration workflow and the strategy that will be applied. [1 mark]

To construct the Smart Shield predictive routing framework, the technology stack is divided into five specialized layers designed to support the ML architecture efficiently:

The team establishes strict version control practices using a standard Git Flow protocol to protect the integrity of the machine learning pipeline and frontend codebase. GitHub link: `https://github.com/1337-KB/Road-Safety-score-app`

- **Repository Management**
  - GitHub serves as the central remote repository hosting code, documentation, and architecture diagrams.
- **Branching Strategy**
  - `main`: Production-ready branch containing stable, tested code.
  - `develop`: Integration branch for finalized features before promotion to main.
  - `feature/*`: Isolated branches for individual development tasks.
- **Collaboration and Review Protocol**
  - Direct pushes to `main` and `develop` are restricted.
  - All code merges require a Pull Request (PR).
  - At least one team member must review and approve each PR before merging.
- **Commit Standards**
  - Commits are pushed regularly with clear and descriptive messages documenting modifications and updates.

Q8: Limitations, Constraints, and Risks

Identify potential limitations, constraints, and risks that may be faced during this AI/ML capstone project. Keep in mind that financial support from Sheridan to teams is not available, and that access this term to GPU supported labs or clusters is not guaranteed. In the space below, develop a risk mitigation table that highlights the top 5 to 7 risks that many encounter while working on this project. The table should also include a contingency plan for each risk. The contingency should propose a backup strategy to mitigate the potential risk, to avoid disruptions to the operations of the workflow, and to allow completion of the AI/ML project. In the table below, provide a heading for each cell of the table that summarizes the risk and the related contingency. Then, provide a description for each cell of the table (in 2 to 3 sentences). [2 marks]

No.	Limitations, Constraints, Risks	Contingency Plan
#1	**Zero Project Financial Budgeting** The team has zero academic or external funding to pay for enterprise APIs or premium infrastructure.	**Utilize Free Tier Alternatives** The team will use free tier options like Open Meteo and free developer accounts from TfL. Production web deployments will use free community hosting solutions.
#2	**No Guaranteed Campus GPU Access** The institution cannot guarantee access to local GPU clusters for training large PyTorch neural networks.	**Leverage Free Cloud Hardware** Free Google Colab notebook allocations will be used to manage model training. Saved model weights will then be exported to run efficiently on standard local CPUs.
#3	**Strict API Ingestion Rate Limits** External transport portals enforce strict hourly request caps that can lock out the pipeline.	**Implement Data Caching Layers** A caching system will be built to store API responses locally for short periods, reducing unnecessary calls during active frontend testing phases.
#4	**Data Format Misalignments** The historical data uses different coordinate layouts and naming systems than the live routing APIs.	**Build a Standardized Mapping Class** Custom translation dictionaries in Pandas will be built to automatically clean up and align different names and coordinate systems into a single format.
#5	**Extreme Real World Weather Changes** Unpredictable shifts in actual weather conditions can lead to sudden gaps in the live test streams.	**Inject Custom Test Scenarios** A collection of mock data payloads representing extreme weather conditions will be created, allowing safe testing of the application anytime regardless of the real weather outside.
#6	**Unstructured Text Processing Failures** Live hazard alerts often contain unexpected phrasing, typos, or chaotic syntax that can confuse the models.	**Implement a Fallback Parsing System** A fallback matching system based on regular expressions will be built to catch known critical keywords if the main model runs into unreadable phrasing.
#7	**Frontend Map Lag and Latency** Processing dense route data on an interactive map can cause the web browser user interface to lag.	**Downsample Route Coordinate Sets** Coordinate reduction techniques will be used to simplify long route lines, keeping the map fast and responsive while maintaining clear safety color coding.

Q9: Project Outcomes

By Week 14, all teams must prepare: (i) D4, a formal technical report detailing the comprehensive work on the AI/ML capstone project, (ii) D5a, slide presentation of the technical report, and (iii) D5b, a working live demonstration of the AI/ML solution. Q9.1 What is the anticipated major outcome of the capstone project? Q9.2 What type of AI/ML method will the solution consider (i.e., regression, classification, clustering, or possibly a combination of different methods)? Q9.3 What type of UI/system will showcase a working demonstration of the AI/ML solution (e.g., MobileApp, cloud service, locally or publicly hosted URL, AI powered physical system like IoT, drone, or robot, or possibly a combination of various UI/systems)? In the space below, provide specific answers to these three questions to demonstrate that the anticipated capstone results are thoroughly considered and strategically planned. [2 marks]

9.1 Anticipated Major Outcome

By Week 14, the Smart-Shield project will deliver a fully realized, intelligent transit safety application designed to reduce road accidents under hazardous conditions. The primary technical outcome will be a functioning, end-to-end multi-modal AI engine that processes live visual and textual environment streams to quantify road segment risk levels. Additionally, the project will yield a comprehensive, publication-grade D4 Technical Report detailing our mathematical fusion methodologies, validation metrics, and architectural decisions. The ultimate tangible deliverable is an accessible, web-based software dashboard suitable for real-world route planning demonstrations.

9.2 AI/ML Method

Our solution implements a comprehensive hybrid ensemble approach combining computer vision classification, natural language processing vectorization, and probabilistic regression optimization.

- **The Vision Brain** uses a deep learning Convolutional Neural Network (CNN) for multi-class classification to categorize physical road hazards from camera frames.
- **The NLP Brain** utilizes TF-IDF text tokenization and keyword statistical weighting to calculate a localized risk index.
- **The Central Fusion Layer** combines these outputs with weather variables using a Logistic Regression model with a SAGA solver, mapping the combined inputs into a final safety probability.

9.3 UI/System

The project will use a publicly hosted web application URL built using the Streamlit framework to showcase the live system demonstration. This interface will feature responsive location input fields, real-time metrics, and clean data readouts. The centerpiece of the user experience will be an interactive map module that tracks routes and displays custom color overlays based on our model's predictions. This web-based deployment ensures that anyone—including faculty examiners—can access and interact with the application from any device without installing local dependencies.

References

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